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Linux Command Line Cheat Sheet by [DaveChild](#)

A cheat sheet of the commands I use most for Linux, with popup links to man pages.

Bash Commands

<u>uname -a</u>	Show system and kernel
<u>head -n1 /etc/issue</u>	Show distribution
<u>mount</u>	Show mounted filesystems
<u>date</u>	Show system date
<u>uptime</u>	Show uptime
<u>whoami</u>	Show your username
<u>man command</u>	Show manual for <i>command</i>

Directory Operations

<u>pwd</u>	Show current directory
<u>mkdir dir</u>	Make directory <i>dir</i>
<u>cd dir</u>	Change directory to <i>dir</i>
cd ..	Go up a directory
<u>ls</u>	List files

ls Options

-a	Show all (including hidden)
-R	Recursive list
-r	Reverse order
-t	Sort by last modified
-S	Sort by file size
-l	Long listing format
-1	One file per line
-	Comma-separated output
-Q	Quoted output

Nano Shortcuts

Files

Ctrl-R	Read file
Ctrl-O	Save file
Ctrl-X	Close file

Cut and Paste

ALT-A	Start marking text
CTRL-K	Cut marked text or line
CTRL-U	Paste text

Navigate File

ALT-/	End of file
CTRL-A	Beginning of line
CTRL-E	End of line
CTRL-C	Show line number
CTRL-_	Go to line number

Search File

CTRL-W	Find
ALT-W	Find next
CTRL-\	Search and replace

More nano info at:

<http://www.nano-editor.org/docs.php>

Bash Shortcuts

CTRL-c	Stop current command
CTRL-z	Sleep program
CTRL-a	Go to start of line
CTRL-e	Go to end of line
CTRL-u	Cut from start of line
CTRL-k	Cut to end of line
CTRL-r	Search history
!!	Repeat last command
! <i>abc</i>	Run last command starting with <i>abc</i>
! <i>abc</i> :p	Print last command starting with <i>abc</i>
!\$	Last argument of previous command
ALT-.	Last argument of previous command
!*	All arguments of previous command
^ <i>abc</i> ^ <i>123</i>	Run previous command, replacing <i>abc</i> with <i>123</i>

Search Files

Screen Shortcuts

screen

Start a screen session.

screen -r

Resume a screen session.

screen -list

Show your current screen sessions.

CTRL-A

Activate commands for screen.

CTRL-A c

Create a new instance of terminal.

CTRL-A n

Go to the next instance of terminal.

CTRL-A p

Go to the previous instance of terminal.

CTRL-A "

Show current instances of terminals.

CTRL-A A

Rename the current instance.

More screen info at:

<http://www.gnu.org/software/screen/>

Bash Variables

env

Show
enviro-
nment
variables

grep -i

echo \$NAME

Output
value of
\$NAME
variable

grep -r

export NAME=value

Set
\$NAME to
value

grep -v

\$PATH

Executable
search
path

grep -o

\$HOME

Home
directory

find /dir/ -name name*

\$SHELL

Current
shell

find /dir/ -user name

find /dir/ -mmin num

whereis command

grep pattern files

Search **File Permissions**

for

pattern in

files

Case

insens

itive

search

Recursive

search

Inverted

search

Show

matched

part of

file only

Find files

starting

with

name in

dir

Find files

owned by

name in

dir

Find files

modified

less than

num

minutes

ago in dir

Find

binary /

source /

manual

for

command

chmod 775 file

Change mode of *file* to
775

chmod -R 600 folder

Recursively chmod
folder to 600

chown user.group file

Change *file* owner to
user and group to *group*

File Permission Numbers

First digit is owner permis-
sion, second is group and
third is everyone.

Calculate permission digits
by adding numbers below.

4 read (r)

2 write (w)

1 execute (x)

IO Redirection

`cmd < file`

Input of `cmd` from `file`

`cmd1 <(cmd2)`

Output of `cmd2` as file input to `cmd1`

`cmd > file`

Standard output (stdout) of `cmd` to `file`

`cmd > /dev/null`

Discard stdout of `cmd`

`cmd >> file`

Append stdout to `file`

`cmd 2> file`

Error output (stderr) of `cmd` to `file`

`cmd 1>&2`

stdout to same place as stderr

`cmd 2>&1`

stderr to same place as stdout

`cmd &> file`

Every output of `cmd` to `file`

`cmd` refers to a command.

Pipes

`cmd1 | cmd2`

stdout of `cmd1` to `cmd2`

`cmd1 |& cmd2`

stderr of `cmd1` to `cmd2`

locate file

Find `file`
(quick search of system index)

File Operations

touch file1

Create `file1`

cat file1 file2

Concatenate files and output

less file1

View and paginate `file1`

file file1

Get type of `file1`

cp file1 file2

Copy `file1` to `file2`

mv file1 file2

Move `file1` to `file2`

rm file1

Delete `file1`

head file1

Show first 10 lines of `file1`

tail file1

Show last 10 lines of `file1`

tail -F file1

Output last lines of `file1` as it changes

Command Lists

`cmd1 ; cmd2`

Run `cmd1` then `cmd2`

`cmd1 && cmd2`

Run `cmd2` if `cmd1` is successful

`cmd1 || cmd2`

Run `cmd2` if `cmd1` is not successful

`cmd &`

Run `cmd` in a subshell

Watch a Command

`watch -n 5 'ntpq -p'`

Issue the 'ntpq -p' command every 5 seconds and display output

Process Management

`ps` Show snapshot of processes

`top` Show real time processes

`kill pid` Kill process with id `pid`

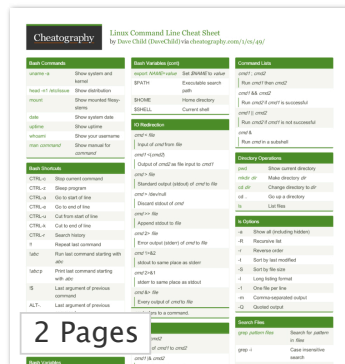
`pkill name` Kill process with name `name`

`killall name` Kill all processes with names beginning `name`

 `bash`  `nano`  `linux`  `ls`  `commandline`  `ubuntu`
 `server`  `sysadmin`  `shell`



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Comments

[a_statham](#), 10:02 28 Nov 11

The I/O redirection section could use "2>" and "&>" examples, I always forget how to redirect stderr

Created By

[DaveChild](#)

<https://aloneonahill.com>

[DaveChild](#), 10:02 28 Nov 11

I've updated that section to include stderr redirection. :)

DaveChild, 10:02 28 Nov 11

Good idea – I'll add that (once I remember how they work myself ... :))

[deleted], 10:02 28 Nov 11

Thanks for creating this cheat-sheet Dave. The one thing I missed was "grep -o"; Show only the part of a matching line that matches PATTERN

DaveChild, 10:02 28 Nov 11

I've added "grep -o" to the cheat sheet :)

wattslevi, 19:51 29 Nov 11

How are the indented lines added several of the cells like Screen Shortcuts? RE:
http://getsatisfaction.com/cheatography/topics/adding_a_two_line_entry_in_a_list

DaveChild, 08:35 30 Nov 11

Those indented bits are a "question and answer" format box.

wattslevi, 10:30 30 Nov 11

Metadata

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By

and 388 more ...

Ah, Q&A means I would be able to cheat with it and use it for two column with an extra line.
Thanks for the info.

(Replying to my original post seems unintuitive for trying to reply under your post, but not to the overall thread... assuming this post as I'm thinking it might.)

jim 23:12 30 Nov 11

On the redirects, the one I most commonly use is ignoring errors (`2>/dev/null`, or more succinctly `2>&-`).

For example, if I'm looking for files and I don't care that I haven't access to parts of the filesystem, we might do something like:

```
find / -name "*.html" 2>&-
```

Tanner 10:30 7 Mar 12

`ls -h` is handy – changes sizes to human readable formats. Goes along good with `-S`. If I'm using it it is generally a `ls -alhS`

Wane 07:28 21 Mar 12

I think the "Bash Shortcuts" part is a little misleading. The "ctrl-a", "ctrl-e", "ctrl-k" is in emacs mode. But there should be many people preferring vim-mode or some thing like that.

Davis Peng 13:57 22 Mar 12

Thanks a lot for your sheet, I just need such a linux command summary such as this sheet.

gamiclea, 00:05 2 Jun 12

Awesome job! This will definitely come in handy

wolvverine 11:29 6 Jun 12

download PDF is broken

Arpit 10:51 29 Jun 12

Good work. Thanks, its helpful.

Chetan Morajkar 14:35 6 Aug 12

Hi,

Fantastic good job..

Thanks

Donald J. Tambeau 15:23 6 Aug 12

Love your work....it is soso helpful! I would like to format a MicroDisk using Linux Centros. When I look at the Partition on the MicroDisk, I see the following comment....(non-Linux). This makes sense, since I

formatted with a computer running Win 7! I want to use it on my Linux system but do not know how to do it. I tried `format /dev/scd1` but no joy!

Thank you

Don

Eric 08:59 7 Aug 12

I'd just like to point out a trick I use with the head and tail commands:

First, you can designate the number of lines to return, like so: `head -100 filename`

I often use both head and tail together to get a section of a file by piping the output of one into the other. The following command gets 100 lines, starting 1000 lines before the end of the file:

`tail -1000 filename | head -100`

You could, of course, reverse the commands to get a section near the top of the file. To get lines 91-100:

`head -100 filename | tail -10`

Mark 08:53 29 Aug 12

I just wanted to take a moment to thank you for putting this together. This is a big help to me I am new to OpenFiler witch I believe uses bash at the core so I am of course new to Linux. I didn't really think it would be quite so difficult to find resources that one can use to navigate the command line but I guess most folks use the GUI. But, I'm not most folks when I bought my first IBM 8088 I started with DOS. Anyway, thanks for the time and effort you put into this, sorry about digressing there.

Shrinath 13:08 4 Oct 12

Excellent Stuff man.. I think these are the most commonly used commands.. Good Effort.

Jimmy 21:43 11 Oct 12

you need to figure out how to split your command sheet as a pdf

Niloufar 21:43 11 Oct 12

hi!.these are useful cheat sheet .

russ 16:18 22 Oct 12

CTRL-Z sleeps (stops) the running process. fg [#] brings it back to foreground.

top

CTRL-Z

ps aux | grep top

fg

chiahsun 11:14 2 Jan 13

Show human readable format (kb, mb...)

ls -lh

PierreBdR, 11:35 9 Feb 13

This cheat-sheet is very good! Thanks for that.

However, I don't like the "chmod" commands you are using. I don't think anybody should use the numeric version of chmod anymore. Your example "chmod -R 600 folder", is the best way to lock yourself out of your own folder and loose any executable bits on the scripts. I would rather use the symbolic version:

```
chmod -R u+rw folder # Add read-write for user on all files in folder
```

```
chmod -R og-rwx # Remove read write and execute bits for "other" and "group" on all files in folder
```

And my all-time favorite:

```
chmod -R og=u-w folder # Give other and group the same rights as user, but removing writing rights.
```

Of course, this also handles t and s bits:

```
chmod u+s file  
chmod o+t folder
```

Stylius 09:17 15 Feb 13

I would also add the tar command. No sysadmin would survive without it.

To extract tar.gz archive

```
tar xvzf archive.tar.gz
```

To extract tar.bz2 archive

```
tar xvjf archive.tar.bz2
```

To extract tar archive

```
tar xvf archive.tar
```

To create archive

```
tar cvzf archive.tar.gz /file_or_folder/to/archive
```

Sandeep 08:43 16 Feb 13

Nice one...really helpful

John 08:43 16 Feb 13

Great resource – thanks for taking the time and trouble to put this out there.

My favorite ls options are '-ltr' . The t sorts files by time, and r reverses that, so newest files show up right above the prompt, no matter how long the listing is.

daveydave400 08:43 16 Feb 13

What about CTRL+y to paste the stuff you cut back in?
How can you leave that out? CTRL+a/e and CTRL+u/y are the pairs I remember.

gymka 08:44 16 Feb 13

Missing: sed
find -exec

Jared 10:33 26 Feb 13

Nice work, but it would be best as a single page PDF.

Darr247 16:31 5 Apr 13

Or if the PDF at least split into 2 pages (so it could be a laminated 2-sided sheet) without cutting commands in half.

PastExpiryDotCom, 18:08 11 Jul 13

This is bash-tastic!

kimcy929, 11:09 9 Aug 13

thank you very much, it is great

Pradeep 15:12 13 Feb 14

There is no commands for shutdown or reboot..

RobertAttfieldDotCom 11:37 14 Feb 14

Good job on the cheatsheet – this will definitely come in handy for my Linux exam next week. Some commands on umask would be a good addition to this cheatsheet :).

Bill 17:30 17 Feb 14

Hi, handy sheet. Just one thing. Find will do a recursive search by default. In the find /dir -name name* should be in double quotes if you want it to recursively find all files starting with name:

```
find /dir/ -name "name*"
```

That's because, without the quotes, the shell will expand the wildcard before handing the parameters to find. To unexpected things might happen if you don't have the quotes. Please see the examples below:

```
$ find .
```

```
.
```

```
./name1
```

```
./name2
```

```
./dir1
```

```
./dir1/name3
```

```
./dir1/name1
```

```
./dir1/test3
```

```
./test1
```

```
$ find . -name name*
```

```
find: paths must precede expression: name2
```

```
Usage: find [-H] [-L] [-P] [-Olevel] [-D
```

```
help|tree|search|stat|rates|opt|exec] [path...] [expression]
```

```
$ find . -name "name*"
```

```
./name1
```

```
./name2
```

```
./dir1/name3
```

```
./dir1/name1
```

```
$ touch dir1/fred1
```

```
$ find .
```

```
.
```

```
./name1
```

```
./name2
```

```
./dir1
```

```
./dir1/name3
```

```
./dir1/name1
```

```
./dir1/test3
```

```
./dir1/fred1
```

```
./test1
```

```
$ find . -name name*
```

```
find: paths must precede expression: name2
```

```
Usage: find [-H] [-L] [-P] [-Olevel] [-D
```

```
help|tree|search|stat|rates|opt|exec] [path...] [expression]
```

```
$ find . -name "name*"
```

```
./name1
```



```
./name2
./dir1/name3
./dir1/name1
$ find . -name fred*
./dir1/fred1
$ find . -name "fred*"
./dir1/fred1
$
```

Edser 17:35 17 Feb 14

grep -B#

This shows what you are searching plus additional lines where number is added. Great for DHCP lease searching.

No4711 09:56 19 Feb 14

I would suggest to put in screen <tty.device> <baudrate> " just in case anybody needs to connect to some serial console"

Henning 09:57 19 Feb 14

Great reference!

The section on Screen is missing "Ctrl-A d" for detaching.

Mads 08:55 26 Feb 14

Great resource – thanks a lot Dave!

Kabir 22:03 10 Mar 14

add task manage comman:
gnome-system-monitor

Isabel Ambriz 13:08 30 Apr 14

Command that might be used to test connectivity of Linux box to the Internet, get logged on user info, and get TCP/IP configuration info. Can anyone help me?

Aaron Tani 08:27 1 May 14

theres also a good one here:
<http://www.techietek.com/2014/04/29/linux-cli-cheat-sheet-wallpaper/>

Damien 10:43 23 Jul 14

I've always found ``mkdir -p path/to/directory`` to be useful.

snakeroot 10:46 14 Jan 15

`head -n1 /etc/issue` as a means to access distribution name isn't reliable, since the post-login screen could have been customized.

A more reliable version would be `sed -nr 's/^\s*PRETTY_NAME=(.*)/\1/p' /etc/os-release`. This should be true even for distros using legacy init (i.e., current Debian, Gentoo and even Slackware).

jt 12:38 22 Jul 15

Is -h? human readable?

DF 08:31 25 Aug 15

would be good to add Ctrl-left/right for jumping arguments, Ctrl-home/end for jumping whole line.

Alex 15:34 28 Sep 15

'echo \$SHELL' actually prints the default shell, 'ps -p \$\$' will print the current shell

John Kershaw 16:04 28 Mar 16

Love the colour, but could it be a slightly darker/stronger green? It comes out almost impossible to read on my colour printer. I tried outputting in greyscale, but the green-on green sections are still unreadable. Needs more contrast.

Alternatively, change all the green texts to black and keep the backgrounds green?

DaveChild, 16:05 28 Mar 16

I agree, it's not a great colour. I've changed it.

Felipe 19:22 3 Apr 16

Could you add "passwd" to the file? :)

GloObi, 15:31 4 Apr 16

thank you for this really useful sheet !

The download button seems to be broken though :(

NatalieMoore, 05:13 10 Apr 16

Hey GloObi, thanks for the heads up about the broken download button on this cheatsheet. We'll get it fixed and I'll let you know when its been done. Have a great day.

NatalieMoore, 06:49 13 Apr 16

Hi GloObi,

This is now fixed. Please let me know if you have any more problems.

Nat

d.falkovsky, 19:54 7 Apr 16

Can't download. Sad.

NatalieMoore, 05:07 10 Apr 16

Thanks d.falkovsky, I will let Dave know the download button on this cheat sheet needs to be fixed. I will let you know once he has fixed it :). Sorry for the inconvenience.

NatalieMoore, 06:48 13 Apr 16

Hi d.falkovsky,

This is now fixed, you should now be able to download. Please let me know if you have any more problems.

Nat

d.falkovsky, 13:15 16 Apr 16

Thank you, Natalie!

ramiissat, 13:01 22 Mar 17

Good job Dave, Thank you :)

tejasshetty.1808, 18:51 8 Jul 17

I think the colour pdf link directs back to the original page

wkthxtn, 21:34 7 Nov 18

Good for quick reference Thank you

hwac121, 06:53 13 Dec 18

All links for download are dead, pleas fix this issue ASAP.
Would be greatly appreciated, thanks.

Chinaguravaiah M, 05:32 13 Sep 19

thank you very much Dave

grantiago, 14:27 17 Jan 21

thanks Dave
besides ^w my most used nano commands:
Alt+Space Go back one word
Ctrl+Space Go forward one word

Sathyanarayanan, 03:44 19 Oct 21

check this sheet!
<https://cheatography.com/sathyanarayanan/cheat-sheets/linux-command-line/>

Shiny Mc Martin, 07:15 20 Jun 24

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 Charles Duarte

 15 Jun 25

 lock, add, init

 1 Page

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 1 Page

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 senseful

 22 Jun 12, updated 13 May 16

 keyboard, notes, organisation, productivity, evernote

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